

Persona2Web: Benchmarking Personalized Web Agents for Contextual Reasoning with User History

Serin Kim¹ Sangam Lee¹ Dongha Lee¹

Abstract

Large language models have advanced web agents, yet current agents lack personalization capabilities. Since users rarely specify every detail of their intent, practical web agents must be able to interpret ambiguous queries by inferring user preferences and contexts. To address this challenge, we present PERSONA2WEB, the first benchmark for evaluating personalized web agents on the real open web, built upon the *clarify-to-personalize* principle, which requires agents to resolve ambiguity based on user history rather than relying on explicit instructions. PERSONA2WEB consists of: (1) user histories that reveal preferences implicitly over long time spans, (2) ambiguous queries that require agents to infer implicit user preferences, and (3) a reasoning-aware evaluation framework that enables fine-grained assessment of personalization. We conduct extensive experiments across various agent architectures, backbone models, history access schemes, and queries with varying ambiguity levels, revealing key challenges in personalized web agent behavior. For reproducibility, our codes and datasets are publicly available at [CODE].

1. Introduction

Large Language models (LLMs) have progressed beyond imitating human reasoning and are beginning to act autonomously on behalf of humans in complex, real-world tasks (Xi et al., 2023). A representative paradigm of such advancement is the web agent, an autonomous system designed to execute multi-step workflows on web environment that require human intervention (Wang et al., 2024). At the core of these capabilities, LLMs enable the agent to interpret

¹Department of Artificial Intelligence, Yonsei University, Seoul, Republic of Korea. Correspondence to: Dongha Lee <donalee@yonsei.ac.kr>.

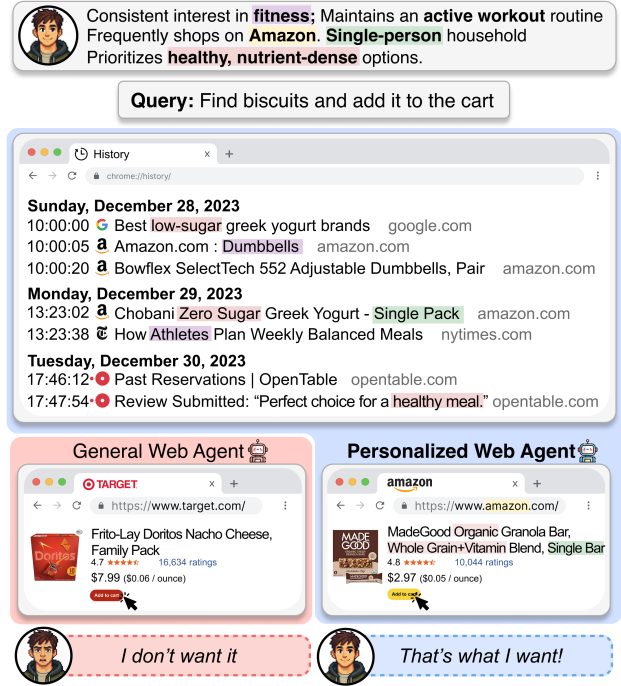


Figure 1. A personalized web agent generates user-specific responses by leveraging user history, whereas a general web agent often produces generic or random outputs that fail to align with user preference.

web contents (Gur et al., 2023; 2024) and understand task-specific instructions (Ouyang et al., 2022), which enhances the robustness of web agents.

However, these advances remain insufficient for fully practical use in real-world scenarios. In real-world scenarios, users rarely specify every detail of their intent, assuming that systems understand their implicit context (Kelly & Teevan, 2003; Hu et al., 2008; Shokouhi et al., 2015; Radlinski & Craswell, 2017; Zamfirescu-Pereira et al., 2023; Deng et al., 2024). To be genuinely practical, web agents must be able to accurately interpret even such ambiguous queries by inferring the user context and preferences. Yet, they often provide generic and sometimes arbitrary responses to ambiguous queries (Song et al., 2007; Qu et al., 2018). In Figure 1, when given the ambiguous query, the general web agent (Left) fails to clarify the query and selects a random high-sugar snack that has no relation to the user’s

preferences. Such behavior yields unsatisfactory outcomes for the user with strong preference for health-conscious options. Personalization can directly address this limitation by enabling agents to leverage user history when resolving ambiguity. When given the same query, the personalized web agent (Right) leverages the user history and selects snacks that align with the user’s health-conscious preferences.

Despite the importance of personalization in web agents, existing benchmarks fail to provide a pertinent evaluation framework for these capabilities. Specifically, they suffer from two fundamental limitations. First, the majority of existing web agent benchmarks do not provide the user context that reveals user’s preference or behavioral patterns which are essential for agents to personalize their responses. Even when some form of context is provided (Zhou et al., 2023; Yao et al., 2025), it does not reflect realistic user behavior that agents would encounter in practice. Second, current benchmarks (Deng et al., 2023; He et al., 2024) rely on fully specified queries that overlook the ambiguity inherent in real user-agent interactions. This eliminates the need for agents to clarify implicit queries based on user context, which is the core challenge of personalization. Consequently, these benchmarks cannot distinguish between agents that genuinely understand and apply user preferences and those that simply follow the instructions.

Motivated by these limitations, we introduce the first personalized web agent benchmark, PERSONA2WEB, specifically designed for evaluating personalization capability of the web agent on the real open web. Central to this benchmark is the principle of *clarify-to-personalize*. This elicits personalization by challenging agents to interpret ambiguous queries and fill in missing details based on user history, rather than simply executing fully specified instructions. Specifically, PERSONA2WEB consists of three core components. **(1) User History:** Our rigorously-constructed user history extends simple browsing logs into richer and more detailed contexts. It reveals user preferences in implicit and indirect ways over long time spans, rather than providing them explicitly at once. **(2) Ambiguous Query:** Our benchmark intentionally conceals specific parts of the query to let the agent clarify based on user history. This design assesses whether the agent can resolve ambiguity without forcing the users to provide exhaustive details. **(3) Reasoning-aware Evaluation:** Beyond simple task completion, our evaluation framework comprehensively examines reasoning traces through structured rubrics to properly distinguish personalization failures from navigation failures. This enables fine-grained assessment of personalization in web agents.

Evaluating both general and personalized web agents on PERSONA2WEB reveals a fundamental gap in current personalization capabilities. Without access to user history, all agents fail completely on ambiguous queries, achieving a

Table 1. Comparison between PERSONA2WEB and existing web agent benchmarks. PERSONA2WEB offers extensive domain and website coverage, enables evaluation on the open web, and supports personalization assessment. Parenthesized numbers next to the environment indicate the number of websites used; simulated, function-call, and dialogue agent therefore omit website counts. Domain counts are recalculated based on our own criteria.

Benchmark	Domains	Environment	Personalization
WebLINX	–	Cached (155)	✗
Mind2Web	31	Cached (137)	✗
WebShop	1	Simulated	✗
WebArena	6	Simulated	✗
WebVoyager	5	Open web (15)	✗
AssistantBench	15+	Open web (258)	✗
WebCanvas	19	Open web (69)	✗
PersonalWAB	1	Function-call	✓
Apollonion	6	Dialogue agent	✓
PERSONA2WEB	21	Open web (105)	✓

0% success rate, which confirms that agents cannot resolve ambiguity without user context. Even when user history is provided, performance improves only marginally, reaching just a 13% success rate at best. These observations indicate that simply supplying user history is insufficient and highlight the need for methods that enable agents to effectively leverage user context for personalization. Furthermore, task completion alone cannot capture personalization capability, as agents with similar success rates excel at personalization but fail at navigation, or the reverse. This underscores the necessity of our reasoning-aware evaluation framework for personalized web agents.

Our key contributions are as follows:

- We propose PERSONA2WEB, the first benchmark for evaluating personalized web agents on the real open web, comprising user histories, personalization query sets, and evaluation framework.
- Our queries follow *clarify-to-personalize* principle that deliberately embeds ambiguity, requiring the agent to infer implicit contexts from user history without explicit cues.
- We conduct extensive experiments across diverse backbone models, history access schemes, and query ambiguity levels to identify key challenges in personalized web agent behavior.

2. Related work

Web agent benchmark. Evaluating the web agents presents inherent challenges due to the dynamic nature of the web. To mitigate this issue, prior benchmarks have primarily relied on simulated or cached environments (Table 1). MiniWOB++ (Shi et al., 2017; Liu et al., 2018) provides simplified HTML-based tasks to test interaction

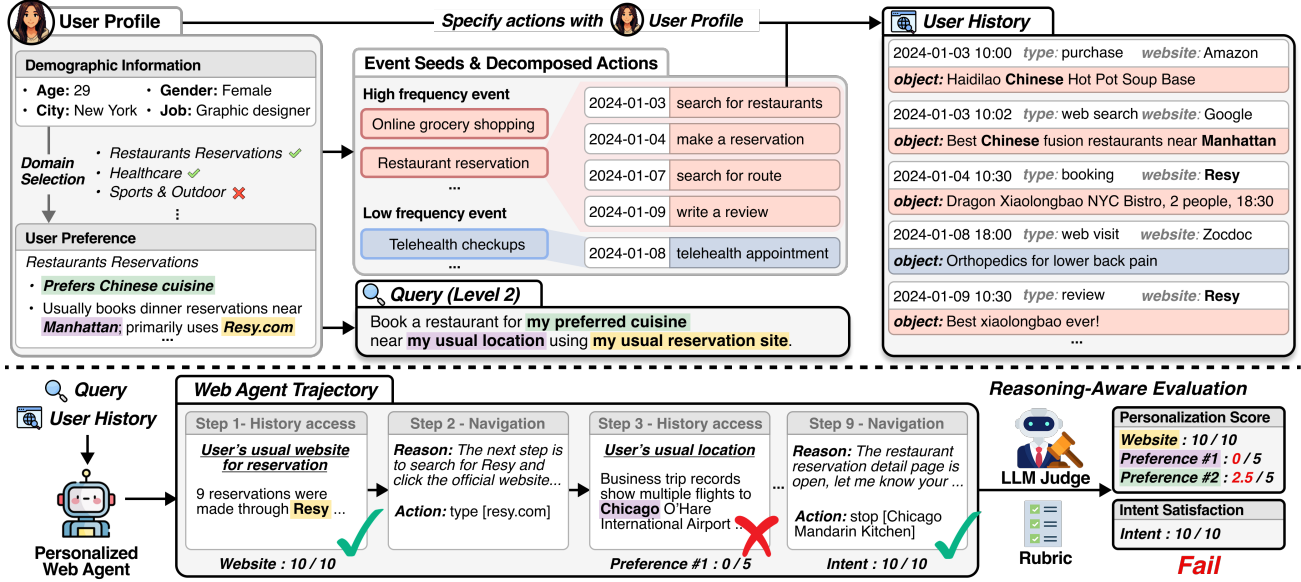


Figure 2. Overview of PERSONA2WEB construction pipeline and reasoning-aware evaluation process.

skills. WebArena (Zhou et al., 2023) introduces a self-hosted simulated platform with end-to-end executable tasks. Mind2Web (Deng et al., 2023) caches real-world webpages and provides realistic tasks demanding multi-step execution. Recent studies have shifted toward benchmarks on the real open web, introducing corresponding evaluation methods to handle its dynamic nature. WebCanvas (Pan et al., 2024) defines mandatory key nodes and evaluates performance at each node, while WebVoyager (He et al., 2024) integrates visual information into observation space and evaluates through LLM judge.

Beyond the environment limitations, existing evaluation methodologies fail to capture the complexity of web agent behavior. Current web agent benchmarks assess performance based on step-wise actions (Gou et al., 2026) or the final outcomes (Yao et al., 2022; Zhou et al., 2023; He et al., 2024) making it impossible to discern whether failures stem from navigation errors, personalization deficiencies, or other reasoning breakdowns. Unlike closed settings, open-web navigation often admits multiple trajectories that lead to correct outcomes, making action-wise or outcome-based evaluation inappropriate.

Personalization in LLMs. Personalization has been explored in memory-augmented LLMs (Kumar et al., 2024; Wu et al., 2025; Pan et al., 2025; Huet et al., 2025) and dialogue agents (Zhang et al., 2018; Xu et al., 2022; Lee et al., 2023; Chen et al., 2024; Zhao et al., 2025a), typically defined as the ability to retrieve and reason over user-relevant information. LaMP (Salemi et al., 2024) introduces a benchmark with user profiles across multiple domains, and LongLaMP (Kumar et al., 2024) extends it with a focus on the text generation tasks. In dialogue settings, LongMemEval (Wu et al., 2025) evaluates memory abilities based

on chat history, Apollonion (Chen et al., 2024) measures personalization through embedding-based similarity with user profiles, and PrefEval (Zhao et al., 2025a) requires models to infer implicitly expressed preferences. PersonalWAB (Cai et al., 2025) explores personalization based on the user’s history, but focuses on abstract function calls rather than web interface interactions. Yet none of these benchmarks address web navigation scenarios, where user preferences are inferred from interaction patterns and apply them in the action planning and execution. Motivated by these limitations, our work moves beyond response generation to encompass realistic web navigation tasks grounded in user context.

3. PERSONA2WEB

In this section, we present PERSONA2WEB, a benchmark designed to evaluate personalized web agents under realistic, open-web environment. It comprises three core components: user history, ambiguous query, and reasoning-aware evaluation. Figure 2 illustrates the overall pipeline.

3.1. Realistic User History

We construct user context in the form of web browsing history, as this format most closely reflects how real web agents would access user information in practice. While dialogue-based benchmarks (Kwak et al., 2025) incorporate user context through explicit verbal cues, browsing history encodes such information implicitly with web behaviors. This requires agents to infer user preferences from actions rather than relying on explicit linguistic cues. To this end, we construct user history through a multi-stage generation pipeline (see Tables 16, 17, 18, and 19 for detailed prompts).

Table 2. Examples of a PERSONA2WEB query set. Each task provides 3 queries with varied ambiguity levels. Web. and Pref. denote website and preferences, respectively. Level 0 is a clear query that includes all explicit cues.

Level	Web.	Pref.	Example Query
0	✓	✓	At zocdoc.com , search for doctors near Southside Jacksonville who is a female .
1	✗	✓	Search for doctors near Southside Jacksonville who is a female at my preferred website.
2	✗	✗	Search for doctors in my usual area that match my preferred provider gender at my preferred website.

The key principle is that user preferences should be *revealed implicitly* from contextual cues distributed across the history, rather than stated explicitly.

User profile generation. The first stage constructs a user profile consisting of *demographic information* $\text{Dem}(u)$ and *domain preferences* $\text{Pref}(u)$. Demographic attributes are evenly distributed to ensure variability across users. Domain preference represents the user’s domain-specific preferences and patterns. We define 21 domains reflecting typical web user experiences.

Given $\text{Dem}(u)$ and the predefined domain set $\mathcal{D} = \{d_1, d_2, \dots, d_{21}\}$, the LLM $\mathcal{M}$ selects K relevant domains with rationales¹:

$$\{(d^{(k)}, \rho^{(k)})\}_{k=1}^K = \mathcal{M}(\text{Dem}(u), \mathcal{D}),$$

where $d^{(k)}$ denotes k -th selected domain and $\rho^{(k)}$ is the rationale describing its relevance. $\mathcal{M}$ then generates domain-specific preferences using domain prompt $\pi^{(k)}$ that specifies what attributes to include in domain k ’s preferences:

$$\text{Pref}(u) = \left\{ \mathcal{M}(d^{(k)}, \pi^{(k)}, \rho^{(k)}, \text{Dem}(u)) \right\}_{k=1}^K$$

The user profile is $\text{Profile}(u) = (\text{Dem}(u), \text{Pref}(u))$ as a result. In total, we construct 50 distinct user profiles.

Event seeds and action decomposition. Since preferences are reflected in event occurrence, randomly generating events without event seed or directly importing external data (Sap et al., 2020) fails to capture consistent preference. We introduce event seeds that define recurring activity patterns based on user preferences and routines. We categorize events into high-frequency (daily activities like commuting, shopping) and low-frequency types (irregular events like travel, medical checkups).

Given the $\text{Dem}(u)$ and the set of rationales within all selected domains $\{\rho^{(k)}\}_{k=1}^K$, LLM generates two event sets: the high-frequency events $\mathcal{E}_u^{\text{HF}}$ and the low-frequency events $\mathcal{E}_u^{\text{LF}}$, as follows.

$$(\mathcal{E}_u^{\text{HF}}, \mathcal{E}_u^{\text{LF}}) = \mathcal{M}(\text{Dem}(u), \{\rho^{(k)}\}_{k=1}^K) \quad (1)$$

We selectively apply cancellations and modifications to approximately 10% of the entire record to reflect the noise and

inconsistencies present in real-world user histories. Each event E_i is decomposed into a sequence of actions:

$$(a_{i,1}, a_{i,2}, \dots, a_{i,L_i}) = \mathcal{M}(E_i), \quad i = 1, \dots, N$$

where $a_{i,j}$ denotes the j -th action derived from event E_i , and L_i is the number of actions in the event E_i . Actions from each event are dispersed across different temporal points over a year. This prevents related actions from being clustered together and instead requires agents to integrate remote histories to recognize recurring patterns.

User history generation. We structure user history with four elements as shown in Figure 2: timestamp, type, object, and website. The *type* field includes five categories, web search, web visit, purchase, booking, and review and rating, capturing a broad spectrum of actions commonly observed in web environments. *Object* provides the detailed information of the target entity for each action type.

3.2. Ambiguous Query Sets

A key distinction of our work lies in intentionally masking explicit values, as the ambiguity of a query should be resolved through the user’s history. Based on this *clarify-to-personalize* approach, we first generate a query with all explicit cues, then create variants by adjusting the level of ambiguity. Each query set consists of three levels (Table 2): **Level 0** (clear query), where both website and preference constraints are explicit; **Level 1**, where preference constraints are explicit while website is masked; and **Level 2**, where both are masked. Level 2 is the target query on which personalized web agents should perform well.

As shown in Tables 3 and 4, PERSONA2WEB provides an extensive, carefully constructed dataset for evaluating personalized web agents across diverse user activities and domains. Notably, only 3.56% of history entries contain preference values as exact string matches. This low ratio attests that user preferences are embedded implicitly through contextual cues rather than as explicit textual signals. Additional validation of user profiles and user histories is conducted in Appendix A.2. The results show that the generated user histories exhibit broad topical coverage, and rich variation. Human evaluation (Table 11) further demonstrates that our multi-stage generation pipeline improves the realism of generated histories over naive generation.

¹We use GPT-4o as $\mathcal{M}$ throughout the data generation pipeline.

Table 3. Overall statistics of PERSONA2WEB. Histories/user, subdomains/user, and websites/user denote averages across user. Domains are reported as domain/subdomain counts. Exact match ratio denotes the average proportion of history entries in which a preference value appears by exact string match.

Statistic	Value	Statistic	Value
Users	50	Queries	150
History entries	102,568	Histories / user	2,051
Domains	7 / 21	Subdomains / user	17
Websites / user	35	Exact match ratio	3.56%

Table 4. Domain coverage of user histories and profiles. Mean values indicate average of history entries and profiles across users.

Domain	User history		User profile	
	Mean	Total	Mean	Total
E-commerce	345	17,272	4	179
Travel/Transportation	659	32,943	4	201
Weather/Maps	271	13,546	1	50
Health/Medical	113	5,670	1	50
Entertainment/Media	261	13,069	3	150
Education/Academia	116	5,816	1	49
Search/Community	299	14,950	2	98

3.3. Reasoning-aware Evaluation

Accurate evaluation of personalized web agents requires distinguishing personalization failures from navigation failures. However, since task completion depends on both personalization and navigation, errors in either are compounded in the final outcome, making the distinction impossible from task success alone. Our evaluation framework examines the full trajectory including reasoning traces based on three metrics: **personalization score**, **intent satisfaction**, and **success rate**. For each metric, we employ scoring rubrics that assign discrete point levels based on whether each criterion is met. To apply these rubrics, we use GPT-5-mini as an LLM judge that receives the agent’s full trajectory along with the rubrics and outputs scores for each metric. Detailed rubrics are provided in Appendix E.

Personalization scores. To evaluate personalization, we define two Personalization Scores (PS), $\mathcal{P}_{\text{web}}$ and $\mathcal{P}_{\text{pref}}$. $\mathcal{P}_{\text{web}}$ determines whether the agent recognizes websites that align with recurring usage patterns based on the user history. $\mathcal{P}_{\text{pref}}$ further examines whether the agent identifies items or attributes that best reflect user preferences. For websites requiring location-aware navigation (e.g., Amazon, Resy), we additionally evaluate whether the agent correctly personalizes based on user location from history, even without an explicit query cue. Since personalization involves multiple steps to retrieve relevant history and utilize it, which can fail independently, each score is computed using two rubrics: *retrieval accuracy*, which assesses whether the agent accesses the correct histories, and *utilization accuracy*, which evaluates whether the retrieved information is appropriately

incorporated into the navigation process. These rubrics require reasoning traces to identify whether failures occur during retrieval or utilization.

Intent satisfaction. Intent satisfaction measures task accuracy apart from personalization. Considering the open-web environment, agents may formulate correct plans but fail due to external factors (website unavailability, dynamic content). The rubric assigns full credit for complete task success and partial credit when the agent attempts the correct action but is blocked by external factors.

Success rate. Success Rate (SR) counts cases with full scores in both PS and IS. The task is deemed successful only when the agent personalizes and executes accurately.

4. Personalized Web Agent

4.1. Architecture

To investigate how well existing web agents can perform personalized web navigation through PERSONA2WEB, we conduct experiments using two web agent architectures, AgentOccam (Yang et al., 2025) and Browser-Use². (For more details about these architectures, see Appendix C.1.) However, these architectures are designed for generic web navigation and do not account for user history by design, making them unsuitable for evaluating personalized web navigation tasks. Therefore, we transform these architectures into the end-to-end personalized web navigation pipelines by augmenting them with a personalization module that retrieves and reasons over user history. Specifically, our transformed architecture consist of three components: **planner**, **retriever**, and **generator**.

- **Planner:** The planner interprets the current observation and determines whether user history is needed. If so, it generates a history query to retrieve relevant history; otherwise, it directly generates the execution plan for the next step.
- **Retriever:** The retriever receives history queries from the planner and searches for relevant entries in the user history. More details in Section C.2.
- **Generator:** The generator reasons over retrieved histories to resolve query ambiguities. It receives the history query from the planner along with the retrieved histories from the retriever and analyzes them to identify preferences and extract specific details. It then produces a personalized rationale that guides subsequent action planning.

Throughout the process, the agent generates a trajectory that captures both actions and reasoning at each step (Erdogan et al., 2025). The resulting reasoning trajectory documents the accessed history entries, query disambiguation process, and final option selection.

²<https://browser-use.com/>

4.2. History Access Scheme

To evaluate distinct aspects of personalization capabilities, we propose two history access schemes for web agents that differ in when and how user history is accessed: **on-demand** and **pre-execution**. In the **on-demand** scheme, the agent accesses user history dynamically during task execution, only when the planner determines it is needed at each step. This scheme evaluates whether the agent can recognize, in real time, when and what type of personalization is required, thereby testing its situational awareness. In the **pre-execution** scheme, the agent retrieves all relevant histories before execution begins by generating multiple queries to resolve ambiguities. This scheme examines the agent’s ability to engage in long-horizon planning to anticipate what information will be needed during execution. Refer to Appendix D for detailed analysis. Full trajectory examples for each scheme are provided in Table 20 and Table 21.

5. Experiments

5.1. Meta Evaluation

To validate our evaluation framework, we conduct a meta-evaluation on 50 newly generated queries by comparing its correlation with human judgments.

Baselines and settings. Using GPT-4.1 as the backbone model, we instruct the agent to perform tasks under both pre-execution and on-demand schemes. GPT-5-mini serves as the meta evaluator, consistent with our main experiment. We compare our method against two other commonly adopted methods: *action-wise*, which evaluates the action sequence across all steps, and *outcome-based*, which considers only the final action and its reasoning. Since these two methods lack access to intermediate history retrieval steps, they evaluate only whether the ground truth preferences are reflected in the agent’s actions when measuring PS. This corresponds to the *utilization accuracy* of our full rubric.

Evaluation process. Human annotators evaluate each pair of trajectory by determining which scheme performs better (pre-execution, on-demand, or tie) across preference, website, and intent, with ground truth values provided for reference. To compare our evaluation framework against human judgment, we convert our scores into pairwise outcomes indicating which scheme received a higher score for each task. We then calculate Pearson and Spearman correlation coefficients, as well as accuracy, between each evaluation method and human annotations across all three metrics to quantify alignment with human judgment. For the reliability of the human-annotated reference, we additionally report inter-annotator agreement in Appendix B.

Evaluation results. In Table 6, our reasoning-aware method achieves the highest correlation with human judg-

ments across all metrics. Notably, ours and the action-wise method both evaluate full trajectories, but ours incorporates reasoning traces. This difference yields a significant performance gap, confirming that reasoning is essential for accurately evaluating personalization. Beyond this alignment with human judgment, the LLM judge’s internal consistency and agreement with alternative judges are further examined in Appendix B.

5.2. Experimental Settings

We assess how personalization performance varies across agent architectures, backbone LLMs, and history access schemes. Specifically, we augment two base architectures (AgentOccam, Browser-Use) with our personalization module (Section 4.1) under three history access schemes (Section 4.2). We evaluate performance using five metrics: $\mathcal{P}_{\text{web}}$, $\mathcal{P}_{\text{pref}}$, $\mathcal{P}_{\text{avg}}$, intent satisfaction, and success rate.

5.3. Main Results

We evaluate personalized web agents with five backbone LLMs: o3 (OpenAI, 2025), GPT-4.1 (OpenAI, 2024), Gemini-2.5-Flash (Google DeepMind, 2024) for proprietary models, Qwen3-80B-Instruct (Qwen Team, 2025), and Llama-3.3-70B-Instruct (Meta AI, 2024) for open-source models. All experiments use level 2 queries, which are the most ambiguous, under three history access schemes. Results are presented in Table 5. We further examine the reproducibility of our results under repeated executions and temporal shifts in Section 5.6.

Performance differs across agent architectures and backbone models. When history access is enabled (on-demand or pre-execution), Browser-Use consistently achieves higher personalization scores ($\mathcal{P}_{\text{web}}$, $\mathcal{P}_{\text{pref}}$) than AgentOccam across all backbone models; this indicates that Browser-Use more effectively applies inferred user preferences during execution. We attribute this gap to differences in observation construction. AgentOccam prunes the accessibility tree to retain only pivotal nodes and their related elements, producing a condensed representation that may discard relevant contexts. Browser-Use, in contrast, constructs a full enhanced DOM tree that preserves richer information (Appendix C.1).

Across both agent architectures, proprietary models (o3 and GPT-4.1) substantially outperform open-source models, achieving success rates of 6–13% under history access, while most open-source models remain below 5%. Among open-source backbones, Qwen3-80B-Instruct is the only model that approaches proprietary-level performance in specific settings. These results suggest that both agent architecture and backbone model capacity jointly constrain personalization performance.

Table 5. Performance comparison across two agent architectures (AgentOccam, Browser Use) with five backbone LLMs under three history access schemes (No-history, On-demand, Pre-execution). Metrics include $\mathcal{P}_{web}$, $\mathcal{P}_{pref}$, $\mathcal{P}_{avg}$ (average of $\mathcal{P}_{web}$ and $\mathcal{P}_{pref}$), Intent satisfaction (Intent), and Success Rate (SR). For No-history scheme, we report only $\mathcal{P}_{avg}$ for brevity.

Model	No-history			On-demand					Pre-execution				
	$\mathcal{P}_{avg}$	Intent	SR	$\mathcal{P}_{web}$	$\mathcal{P}_{pref}$	$\mathcal{P}_{avg}$	Intent	SR	$\mathcal{P}_{web}$	$\mathcal{P}_{pref}$	$\mathcal{P}_{avg}$	Intent	SR
AgentOccam													
o3	0.168	0.503	0.00	0.747	0.551	0.649	0.423	0.07	0.683	0.555	0.619	0.397	0.07
GPT-4.1	0.126	0.503	0.00	0.740	0.577	0.658	0.417	0.06	0.705	0.574	0.639	0.409	0.07
Gemini-2.5-Flash	0.118	0.500	0.00	0.603	0.368	0.486	0.403	0.02	0.517	0.208	0.362	0.377	0.00
Qwen3-80B-Inst.	0.122	0.463	0.00	0.620	0.480	0.550	0.453	0.07	0.703	0.646	0.675	0.347	0.05
Llama-3.3-70B	0.130	0.242	0.00	0.683	0.491	0.587	0.140	0.01	0.693	0.552	0.622	0.217	0.00
Browser-Use													
o3	0.034	0.277	0.00	0.743	0.568	0.655	0.530	0.13	0.733	0.608	0.671	0.450	0.10
GPT-4.1	0.077	0.250	0.00	0.803	0.731	0.767	0.467	0.13	0.767	0.686	0.727	0.530	0.13
Gemini-2.5-Flash	0.095	0.270	0.00	0.707	0.487	0.597	0.283	0.02	0.757	0.561	0.659	0.350	0.03
Qwen3-80B-Inst.	0.098	0.230	0.00	0.797	0.551	0.674	0.297	0.03	0.760	0.680	0.720	0.407	0.03
Llama-3.3-70B	0.078	0.090	0.00	0.690	0.534	0.612	0.190	0.01	0.713	0.647	0.680	0.197	0.02

Table 6. Meta evaluation results comparing reasoning-aware (ours), action-wise, and outcome-based evaluation.

Method	Eval metric	Pearson	Spearman	Accuracy
Reasoning-aware (ours)	Website	0.8525	0.8611	0.8400
	Preference	0.7200	0.7248	0.8800
	Intent	0.7386	0.7436	0.7400
Action-wise	Website	0.7654	0.8163	0.8400
	Preference	0.3993	0.4026	0.5600
	Intent	0.6409	0.6330	0.6800
Outcome-based	Website	0.8038	0.8209	0.8300
	Preference	0.2253	0.2241	0.4600
	Intent	0.5909	0.5933	0.7000

Web agents cannot handle ambiguous queries without user history. Under no-history scheme, all models across both agents achieve 0% success. This confirms that web agents cannot resolve ambiguous queries without access to user history. Notably, the two agents exhibit distinct failure patterns. AgentOccam achieves approximately $2\times$ higher intent satisfaction than Browser-Use by substituting missing values with random values, prioritizing task completion over accuracy. In contrast, Browser-Use requests user clarification and terminates execution, prioritizing accuracy over completion but resulting in task abandonment. In both cases, ambiguity remains unresolved. This inability to handle ambiguity cannot be captured through existing benchmarks that provide explicit queries, which highlights the necessity of our *clarify-to-personalize* queries and user history.

Task completion alone cannot measure personalization capability. For personalized web agents, task success depends on both personalization and navigation, and weakness in either leads to failure. For instance, Browser-Use with Llama-3.3-70B under pre-execution achieves $\mathcal{P}_{avg} = 0.680$ but only Intent = 0.197, which results in SR = 0.02. Conversely, AgentOccam with Gemini-2.5-Flash under on-demand achieves Intent = 0.403 but only $\mathcal{P}_{avg} = 0.486$,

which also results in SR = 0.02. Both cases record the same low success rate, yet they fail for opposite reasons. Llama succeeds at personalization but struggles with navigation, while Gemini navigates reasonably but fails at personalization. That is, task completion alone cannot distinguish whether failures arise from incorrect personalization or ineffective navigation. Our reasoning-aware evaluation framework resolves this by evaluating personalization and navigation separately.

History access schemes reveal backbone strengths. Some backbone models consistently perform better under a specific scheme regardless of agent architecture. For instance, GPT-4.1 consistently achieves higher $\mathcal{P}_{avg}$ under on-demand than pre-execution across both agents, while Qwen3-80B-Instruct and Llama-3.3-70B show the opposite pattern, performing relatively better under pre-execution. This indicates that the two schemes capture different capabilities of these models as mentioned in Section 4.2. Based on this result, GPT-4.1 appears better suited for situational awareness, while Qwen3-80B-Instruct and Llama-3.3-70B are for long-horizon planning.

5.4. Analysis Across Query Ambiguity

This section analyzes how performance changes across query ambiguity levels for each combination of agent architecture, backbone models, and two history access schemes. Based on the main results presented in 5.3, we select the top three performing backbone models—o3, GPT-4.1, and Qwen3-80B-Instruct—for the analysis. Figure 3 reports success rate and personalization scores across different ambiguity levels (level 0–2).

Performance degrades as ambiguity level increases. Well-personalized web agents should maintain performance even on ambiguous queries by leveraging user history. How-

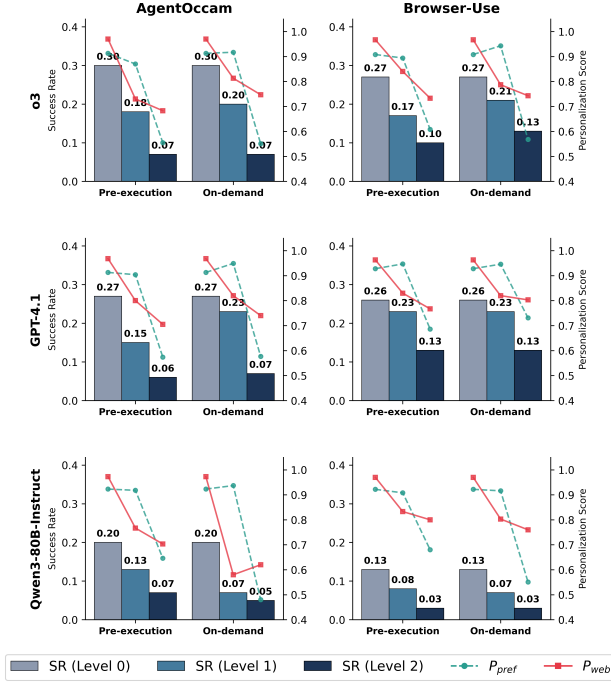


Figure 3. Performance across query ambiguity levels for each agent architecture and history access scheme. Bars show Success Rate (left axis) for level 0, 1, 2 (gray, medium, dark blue). Lines show preference (red) and website (green) scores (right axis). o3, GPT-4.1, and Qwen3-80B-Instruct are used as backbone models.

ever, despite providing access to user history through our personalization module, success rate consistently decreases as query ambiguity increases across all configurations. Clear queries achieve an average success rate of 23.8%, while level 1 queries drop to 16.3%, and level 2 to 7.8%. Since both the personalized architecture and user history are available, this performance drop suggests that agents struggle to effectively identify, reason over, and utilize relevant historical information to disambiguate queries as ambiguity increases.

Clear queries still lead to personalization failures. To identify factors limiting personalization capability, we analyze performance on clear queries, where all required information is explicitly provided and no retrieval is needed. Yet, preference scores average only 0.92 and website scores 0.97 on clear queries, neither reaching a perfect 1.0. This indicates that even when complete information is given, agents often fail at utilization. We trace these failures to two primary causes. First, agents struggle to handle multiple constraints, applying only a subset. For example, given “my usual external SSD capacity, port, and budget,” agents correctly applied capacity (1TB-1.9TB) and port (USB interface) filters, but failed to apply the budget constraint (under \$120). Second, agents fail to determine when and where to apply given information, specifically identifying the appropriate element or execution step remains challenging. These findings indicate that improving personalization requires

Table 7. Performance with explicit user profiles using Browser-Use. Values in parentheses indicate changes relative to implicit user history results in Table 5.

History Scheme	Metric	Backbone	
		o3	GPT-4.1
Pre-execution	$\mathcal{P}_{web}$	0.830 (+0.097)	0.887 (+0.120)
	$\mathcal{P}_{pref}$	0.764 (+0.156)	0.892 (+0.206)
	Intent	0.400 (+0.050)	0.640 (+0.110)
	Success	0.17 (+0.07)	0.27 (+0.14)
On-demand	$\mathcal{P}_{web}$	0.890 (+0.147)	0.933 (+0.130)
	$\mathcal{P}_{pref}$	0.682 (+0.114)	0.887 (+0.156)
	Intent	0.617 (+0.087)	0.597 (+0.130)
	Success	0.15 (+0.02)	0.25 (+0.12)

Table 8. Repeated experiments over three runs using Browser-Use.

Model	Metric	On-demand	Pre-execution
GPT-4.1	$\mathcal{P}_{web}$	0.834 ± 0.015	0.807 ± 0.013
	$\mathcal{P}_{pref}$	0.790 ± 0.005	0.790 ± 0.006
	Intent	0.643 ± 0.025	0.559 ± 0.015
	Success	0.022 ± 0.000	0.018 ± 0.003
Qwen-80B	$\mathcal{P}_{web}$	0.779 ± 0.005	0.807 ± 0.007
	$\mathcal{P}_{pref}$	0.560 ± 0.003	0.774 ± 0.008
	Intent	0.301 ± 0.012	0.448 ± 0.025
	Success	0.008 ± 0.001	0.012 ± 0.001

advancing not only retrieval but also utilization capabilities.

5.5. Analysis On User Context Implicitness

We construct our user history to reveal preferences implicitly through behavioral patterns rather than explicit statements. This is critical since explicit statements reduce personalization to instruction-following, where agents simply apply given information, not inferring user intent. This undermines the genuine evaluation of personalization. To validate our implicit encoding and its necessity for personalization evaluation, we compare performance when given **explicit user profile** (Section 3.1), which explicitly includes preference statements, and **implicit user history**, which implicitly reveals preferences via behavioral patterns. In Table 7, all configurations achieve substantially higher performance across all metrics with explicit profiles. Both contexts contain the same user information, yet there exists significant performance gap. This confirms that preferences in our user history are not explicitly visible but must be inferred from behavioral patterns, validating our implicit encoding design. Furthermore, it highlights the necessity of implicit user context for genuine personalization and evaluation.

5.6. Reproducibility under Open-web Dynamics

Since agents on the open web encounter dynamic environments (e.g., different search results, item availability) across executions (Pan et al., 2024; Rawles et al., 2025; He et al., 2024), reliable evaluation must distinguish agent behavior

Table 9. Temporal variance between December 2025 and March 2026 runs using Browser-Use.

Model	Scheme	$\mathcal{P}_{\text{web}}$	$\mathcal{P}_{\text{pref}}$	Intent	Success
GPT-4.1	On	+0.031	+0.059	+0.176	-0.108
	Pre	+0.040	+0.104	+0.029	-0.112
Qwen-80B	On	-0.018	+0.009	+0.004	-0.022
	Pre	+0.047	+0.094	+0.041	-0.018

from such variability. We therefore examine whether PERSONA2WEB yields stable scores under open-web dynamics in two dimensions: execution variability and temporal variability. Execution variability measures score variation across repeated runs under similar time conditions, while temporal variability measures variation across runs collected several months apart. We conduct both analyses on a representative subset, using Browser-Use with the best performing proprietary and open-source backbones in Table 5, GPT-4.1 and Qwen-80B-Instruct, respectively.

Execution variability Table 8 shows that three independent runs produce low variance, with a maximum standard deviation of 0.025. This indicates stable scoring under repeated open-web execution.

Temporal variability Table 9 reports the differences between the scores in Table 5 and 8, obtained three months apart. Most metrics exhibit small absolute differences between the two periods. The relative model ranking also remains unchanged over time, as GPT-4.1 continues to outperform Qwen-80B-Instruct across schemes and metrics.

Overall, these findings indicate that the results in Table 5 remain consistent despite open-web dynamics, supporting the faithfulness of the evaluation. Our evaluation pipeline is based on reasoning traces rather than final outcomes, and allows partial credit when external factors block correct actions that would otherwise succeed. This prevents open-web variability from being conflated with agent capability.

6. Error Analysis

To investigate the sources of personalization failures, we conduct an analysis based on the agent’s trajectories from level 2 query experiments. We examine them with imperfect PS and categorize observed errors into four distinct types: Redundant history access, Personalization hallucination, History retrieval failure, and History utilization failure. Beyond the qualitative analysis, Appendix I provides complementary error statistics across backbone models.

Redundant history access. Redundant history access occurs when the agent generates underspecified personalization queries, leading to unsuccessful retrieval and repeated attempts. In Table 22, the agent queries “*user’s usual shore excursion preferences*,” which is too broad to identify rele-

vant history. When the query is refined by including domain-aware attributes “*activity type, activity level, price range*”, the agent retrieves relevant history. Without such specificity, the agent repeatedly accesses user history assuming no relevant history exists, causing redundant retrieval attempts.

Personalization hallucination. Personalization hallucination refers to errors in which the model fabricates user information without accessing relevant histories. In Table 23, the agent incorrectly applies a non-existent preference *4 stars and up* as if derived from the user history. Such errors frequently occur with rating or pricing thresholds, where the agent defaults to generic assumptions (e.g., preferring higher ratings or lower prices) instead of referencing user history. Although such behavior may appear operationally sound, it contradicts the goal of personalization. The goal is not to generate plausible information, but to reason faithfully from user-grounded evidence.

History retrieval failure. History retrieval failure refers to cases in which the agent retrieves irrelevant or incomplete histories, leading to the omission of essential information. In Table 24, the agent fails to identify “*Baltimore Orioles*” as the user’s favorite MLB team. Although it is mentioned in multiple histories, none of them explicitly state “favorite team” causing the retriever to miss the connection. To understand whether retriever design contributes to such failures, we explore alternative indexing designs in Appendix F.

History utilization failure. History utilization failure represents situations where appropriate history is successfully retrieved but not properly utilized in the agent’s action. In Table 25, the agent retrieves the user’s location *Los Angeles* but fails to incorporate it into searching, resulting in incorrect results. This failure stems from the agent’s inability to determine when the retrieved histories should be applied within the overall process. The location needs to be set prior to initiating the search to yield accurate results.

7. Conclusion

This paper establishes the first benchmark for evaluating personalized web agents in real open web environments. Our benchmark introduces a *clarify-to-personalize* query design and rigorously constructs user histories, along with a reasoning-aware evaluation framework that assesses personalization through rubric-based scoring. We conduct extensive experiments across web agent architectures, multiple backbone LLMs, two history access schemes, query ambiguity, and user context formats, followed by error analysis to identify key challenges in web agent personalization. By enabling the assessment of how agents infer and apply user preferences, we believe this benchmark provides a strong foundation for advancing personalization in web agents.

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Impact Statement

This paper presents work whose goal is to advance the field of personalized web agents. Our PERSONA2WEB uses synthetically generated user profiles and browsing histories, ensuring no real user data is collected or used. However, we acknowledge that deployed personalized agents would require access to actual user behavior patterns, raising privacy considerations regarding consent and data protection. We encourage future work to carefully address these concerns. Additionally, as our benchmark operates on the real open web, agents may encounter potentially harmful content during execution. We hope our work encourages research into personalization approaches that incorporate both privacy safeguards and safety mechanisms for web navigation.

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A. Dataset

A.1. Dataset Details

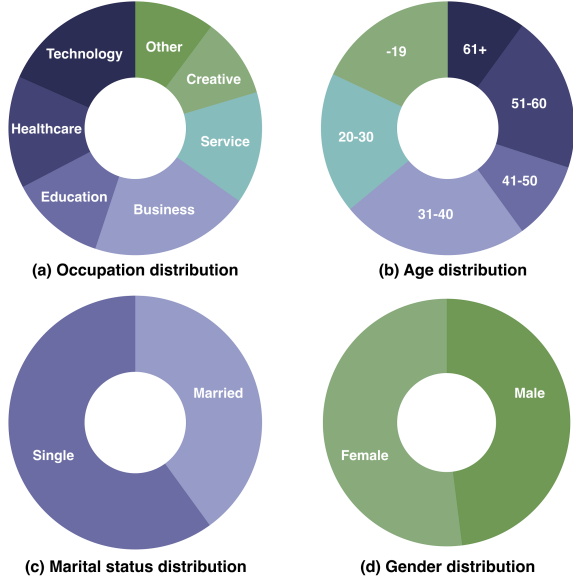


Figure 4. Detailed statistics of user profile provided from PERSONA2WEB benchmark.

We construct a user profile of 50 individuals whose occupation, age, gender, and marital status are evenly distributed to ensure demographic balance, as depicted in Figure 4. Each attribute combination is designed to be plausible and internally consistent, avoiding unrealistic pairings such as a 16 years old professor or a 21 years old retiree. To ensure this, both human validation and prompt-based filtering are employed during generation. The occupation categories are defined across six major sectors—technology, healthcare, education, business, service, and creative industries. To define realistic web domains, we collected 144 websites used in prior benchmarks (Mind2Web, WebArena, WebVoyager) and filtered out sites with CAPTCHAs, regional restrictions, or outdated content, resulting in 105 accessible websites. We categorized these into 21 representative domains: Groceries, Apparel and accessories, Electronics, Home and kitchen, Books and stationary, Sports and outdoor, Transportation, Hotels, Car rental, Tours, Restaurants reservations, Weather and maps, Health and medical, Movies, Music, Concerts and exhibitions, Games, Online learning, Search, and Communities and forums. The inclusion of diverse occupations and domains results in heterogeneous user profiles, each exhibiting distinct interests and behavioral routines, thereby enabling the generation of diverse and realistic user interaction histories. We construct approximately 2,000 history entries spanning a one-year period for each of the 50 users.

A.2. Dataset Validation

User Profile We employed GPT-5 during data construction (Zhao et al., 2025a; Kwak et al., 2025) to verify whether all the generated event seeds and decomposed actions adhere to the generation instructions (Table 8-10) and to detect errors (e.g., formatting error, temporal inconsistency, preference misalignment, etc). All identified issues were corrected before generating the user histories.

Human annotators additionally reviewed each user profile against five criteria: (a) plausibility and internal consistency of demographic field combinations, (b) compatibility between preferences/behavioral patterns and demographic characteristics, (c) whether selected websites actually support the described functionalities, (d) whether behavioral patterns exhibit incoherent mixtures of unrelated routines, and (e) absence of conflicts between preferences and behavioral patterns. Profiles failing any criterion were revised and re-examined until all were satisfied.

Table 10. Diversity of user histories across global, cross-user, and per-user scales.

Scale	Metric	Value
Global	Topic entropy	4.503
	Shannon evenness	0.978
Cross-user	Cosine distance	0.626
	Self-BLEU	0.235
Per-user	Ent-4	9.932

Diversity of User History PERSONA2WEB’s user histories must be sufficiently diverse to reveal distinct preferences across users and rich behavioral patterns within each user. We therefore validate user history diversity at three scales: global (overall topical coverage), cross-user (inter-user distinctiveness), and per-user (intra-user variation). In addition to standard lexical diversity metrics (Ent-4, Self-BLEU, Shannon evenness) used in [Joko et al. \(2024\)](#); [Zhao et al. \(2025b\)](#), topic clustering over sentence embeddings are adopted to capture semantic diversity beyond lexical variation. Results are reported in Table 10.

At the global scale, topic entropy and Shannon evenness measure how broadly and uniformly history entries from all users span topic clusters obtained from sentence embeddings. The topic entropy of 4.503 (out of a maximum of 4.605) and an evenness of 0.978 together indicate broad coverage without any dominant subset. At the cross-user scale, the cosine distance between users’ topic distributions and the Self-BLEU between users’ entries quantify topical and textual distinctiveness. The result attains 0.626 and 0.235 respectively, proposing the contextual distinctiveness with low textual overlap. At the per-user scale, Ent-4 measures the entropy of 4-gram distributions within each individual

history. The Ent-4 of 9.932 exceeds the prior leading score of 8.72 (Zhao et al., 2025b), which demonstrates the high lexical diversity within each user’s history.

Table 11. Pairwise human evaluation of user history realism. Win Rate denotes the proportion of judgments where PERSONA2WEB is preferred over histories generated without our pipeline. Gwet’s AC1 measures IAA among annotators.

Level	Win Rate	Gwet’s AC1
History sequence	88.3%	0.72
History entry	98.3%	0.97

Realism of User History To assess whether the generated histories reflect plausible human behavioral patterns, we conducted a pairwise human evaluation against histories generated without our multi-stage pipeline. Six human judges were provided with the pair of each profile and asked to select which one more faithfully represents the web activity of the given user. The evaluation covers two dimensions. The first, *history sequence*, examines whether the action sequence reflects realistic behavioral flow. The second, *history entry*, captures how clearly individual entries manifest user-specific preferences. Our dataset achieved win rates of 88.3% and 98.3% at the sequence level and entry level, respectively. Furthermore, human inter-annotation agreement (IAA) measured by Gwet’s AC1 reached 0.72 and 0.97 respectively, indicating strong agreement.

B. LLM Judge Validation

B.1. Human Inter-annotator Agreement

Table 12. Human inter-annotator agreement in meta-evaluation.

Metric	$\mathcal{P}_{\text{web}}$	$\mathcal{P}_{\text{pref}}$	Intent
Average agreement	100.0	92.0	81.3
Gwet’s AC1	1.000	0.902	0.743

In our meta-evaluation (Section 5.1), human judgments serve as the reference for evaluating how well each evaluation method aligns with humans. To verify that the human reference is reliable, three annotators independently label the same set of trajectory pairs and measure IAA on each metric. We report both average agreement, which measures the proportion of agreement between annotators, and Gwet’s AC1, which adjusts this agreement for chance. As shown in Table 12, annotators reach perfect agreement on $\mathcal{P}_{\text{web}}$, near perfect agreement on $\mathcal{P}_{\text{pref}}$ (AC1 = 0.902), and substantial agreement on intent (AC1 = 0.743). This supports the faithfulness of human judgments as a reference for assessing our evaluation framework.

Table 13. Average and standard deviation across three repeated evaluations of GPT-5-mini on Browser-Use with GPT-4.1 trajectories.

Scheme	$\mathcal{P}_{\text{web}}$	$\mathcal{P}_{\text{pref}}$	Intent
On	0.828 ± 0.006	0.788 ± 0.003	0.619 ± 0.002
Pre	0.811 ± 0.006	0.771 ± 0.012	0.556 ± 0.015

B.2. Internal Consistency

We validate that our LLM judge (GPT-5-mini) produces consistent scores by evaluating the same trajectories (Browser-Use with GPT-4.1) three times and computing the variance of each metric across the three evaluations. The success rate is excluded since it is computed rule-based. Table 13 presents stable scores with standard deviations no greater than 0.015 across all metrics and history access schemes, confirming the reliability of our LLM judge.

B.3. Inter-judge Agreement

Table 14. Inter-judge agreement across three LLM judges (GPT-5-mini, Qwen-80B-Instruct, GPT-OSS-120B) on Browser-Use with GPT-4.1 trajectories.

Agreement	Scheme	$\mathcal{P}_{\text{web}}$	$\mathcal{P}_{\text{pref}}$	Intent
Krippendorff’s α	Pre	0.837	0.788	0.676
	On	0.801	0.735	0.615
ICC(A,1)	Pre	0.839	0.789	0.679
	On	0.802	0.737	0.608

Beyond internal consistency, we additionally examine the robustness to the choice of judge models. Table 14 reports the agreement across three judges: GPT-5-mini, Qwen-80B-Instruct, and GPT-OSS-120B. Two open-source judges are intentionally included to mitigate potential bias from relying solely on proprietary models. To reduce variance, each score is calculated on average of three runs on the same trajectories (Browser-Use with GPT-4.1). Krippendorff’s α (Chaudhary et al., 2024; Haldar & Hockenmaier, 2025) and ICC(A,1) (Li et al., 2024) are reported, interpreted following Krippendorff’s scale and the guideline of Koo & Li (2016), respectively. $\mathcal{P}_{\text{web}}$ and $\mathcal{P}_{\text{pref}}$ reach the reliable threshold ($\alpha \geq 0.800$) and the excellent ICC level (≥ 0.75) in most conditions, while intent remains within the tentatively acceptable range. These results demonstrate that our evaluation framework is robust to the choice of judge model.

C. Implementation Details

C.1. Web Agent

AgentOccam AgentOccam (Yang et al., 2025) improves web navigation performance without additional training by simply aligning the observation and action spaces with the LLM’s inherent capabilities. It simplifies action space to

reduce distractions and improving focus on meaningful operations. For the observation space, it removes redundant textual elements that describe page layout or functionality to produce a condensed representation of each page. It then refines observation history selectively by identifying pivotal nodes on the accessibility tree, which are web elements essential for task completion, and retaining these nodes along with their ancestors, siblings, and descendants. It then organizes the workflow through a planning tree to keep the agent focused. Each branch represents a new sub-goal and once a branch is initiated, past trajectories irrelevant to the current plan are pruned. Following AgentOccam, we use a Playwright-based environment with Chromium as the browser engine. Unlike the original AgentOccam setup, we set the starting page to `bing.com` to avoid CAPTCHA issues.

Browser-Use Browser-Use extracts web page state through the Chrome DevTools Protocol (CDP). It retrieves three types of information in parallel: the DOM tree, layout information (bounding boxes and computed CSS styles), and the accessibility tree containing semantic role information for each element. These three data sources are merged into an enhanced DOM tree where each node contains its structural properties, screen coordinates, visibility status, scrollability, and accessibility semantics. The framework then serializes this enhanced tree into a text representation for LLM input, assigning numeric indices to interactive elements. Browser-Use automatically switches to `duckduckgo.com` whenever it encounters CAPTCHAs.

While AgentOccam selectively retains pivotal nodes and their related elements to produce a condensed observation, Browser-Use constructs the full DOM tree and filters elements based on visibility and interactivity derived from the CDP snapshot data. For fair comparison, we use only text-based observations without screenshot images.

C.2. Retriever

As a retriever, we use Stella V5 1.5B, the dense retriever that has shown high performance on MTEB (Muennighoff et al., 2023). Each history entry is indexed using the concatenation of its *type* and *object* attributes as the key, while the complete history item is stored as the value. We employ FAISS with inner product similarity (IndexFlatIP) for efficient nearest neighbor search. At retrieval time, we compute cosine similarity between the query embedding and all indexed keys, returning up to 20 entries that exceed a similarity threshold of 0.5. The embedding cache and FAISS index are serialized and stored on disk to enable fast loading across multiple task executions.

D. History Access Schemes

Our history access schemes evaluate fundamentally different dimensions of personalized web agent.

In the **on-demand** scheme, our analysis targets the agent’s capacity to detect *when* personalization is required and *what* kind of user information is needed based on the current state, not compensating for missing inputs with generic or randomly imputed values. The focus is thus not merely on end-to-end completion, but on whether the agent adheres to the *clarify-to-personalize* goal, which is demonstrated by requesting user context via personalization module precisely when the current step demands it. In practice, web agents cannot assume users are available to answer queries at every step of execution. In such environments, well-personalized web agents must be able to proactively access to the history when it is needed. On-demand evaluation directly probes real-time situational awareness and the necessity of timely personalization intervention under realistic constraints.

Pre-execution requires the agent to *anticipate* what user information will be needed based on the given intent and domain, then *integrate* retrieved histories to construct an enhanced query that resolves ambiguities. Anticipation demands long-horizon planning, while integration requires accurately synthesizing information across multiple domains. Unlike the on-demand scheme, history access is unavailable during navigation. Thus, any omission of constraints, inclusion of irrelevant histories, or reliance on stale information propagates throughout the entire trajectory.

E. Evaluation Rubric

We employ GPT-5-mini to evaluate agent performance. The model receives the agent’s full trajectory, including reasoning traces, and measures personalization and intent scores based on the rubrics. Below, we detail the scoring criteria for each metric.

E.1. Website Score

The website score $\mathcal{P}_{\text{web}}$ evaluates whether the agent correctly identifies and navigates to the user’s preferred website based on user history. Total score: 10 points (5 for retrieval + 5 for utilization).

Website Personalization Rubric

Task Context: The agent needed to identify and navigate to the correct website based on the task query.

STEP 1 - Retrieval Evaluation:

1. Did the agent generate appropriate queries to find website preference/history?
2. Was the expected website found in ANY of the memory responses?

STEP 2 - Utilization Evaluation:

1. Did the agent successfully navigate to the expected website?

Scoring:

- **STEP 1 - Retrieval:** 5 pts if criteria met, 0 otherwise
- **STEP 2 - Utilization:** 5 pts if criteria met, 0 otherwise

E.2. Preference Score

The preference score $\mathcal{P}_{\text{pref}}$ evaluates whether the agent correctly retrieves and utilizes user preferences. Total score: 10 points (2.5 per preference $\times$ 2 preferences $\times$ 2 steps). For location-aware websites, an additional 5 points for location personalization. (Total score: 15 points) The total score is normalized: $\mathcal{P}_{\text{web}} = \frac{\text{raw score}}{\text{max score}}$.

Preference Personalization Rubric

Task Context: The agent needed to retrieve and apply user preferences for the given query.

Target Preferences:

- Preference 1 - Category: {pref1_key}, Expected Value: {pref1_value}
- Preference 2 - Category: {pref2_key}, Expected Value: {pref2_value}

STEP 1 - Retrieval Evaluation:

For EACH preference:

1. Check if any memory query asked about this preference category
2. Check if the expected value appears in any memory response

STEP 2 - Utilization Evaluation:

For EACH preference:

1. Check if the expected value was correctly applied in the reasoning
2. OR check if the agent at least attempted to apply the expected value

Scoring:

For EACH preference:

- **STEP 1 - Retrieval:** 2.5 pts if criteria met, 0 otherwise
- **STEP 2 - Utilization:** 2.5 pts if criteria met, 0 otherwise

E.2.1. LOCATION EVALUATION (CONDITIONAL)

For websites requiring location-aware navigation (Amazon, Instacart, Resy, OpenTable, etc), we additionally evaluate location personalization using similar retrieval and utilization rubrics, each worth 2.5 points. The total score is normalized:

$$\mathcal{P}_{\text{pref}} = \frac{\text{raw score}}{\text{max score}}$$

Location Personalization Rubric

Task Context: The agent visited a location-required website and needed to apply user location.

Expected Location: {city, state, zip_code}

STEP 1 - Retrieval Evaluation:

1. Did the agent query for location/address information from memory?
2. Was location data (city, state, zip code, address, neighborhood) retrieved?

STEP 2 - Utilization Evaluation:

1. Was the location applied during navigation? (delivery address, search location, filters, etc.)

Scoring:

- **STEP 1 - Retrieval:** 2.5 pts if criteria met, 0 otherwise
- **STEP 2 - Utilization:** 2.5 pts if criteria met, 0 otherwise

E.3. Intent Satisfaction

Evaluates task completion independent of personalization accuracy, with partial credit for external failures. Total score: 10 points.

Intent Satisfaction Rubric
10 POINTS - Complete Intent Satisfaction:

- The requested final goal was successfully achieved
- Examples: item added to cart, booking completed, information found

5 POINTS - Critical Action Attempted but Blocked:

- Agent reached the correct website/service
- Agent identified the correct item/service
- Agent attempted the final decisive action (e.g., clicked "Add to Cart")
- BUT was blocked by external factors: out of stock, CAPTCHA, website error, service unavailable

0 POINTS - Intent Not Satisfied:

- Failed to reach the decisive action
- Search/filtering failed
- Wrong item selected
- Process abandoned
- Constraint violated (e.g., attempted login when prohibited)

E.4. Success Rate

A task is considered successful if and only if all three metrics achieve perfect scores:

F. Effect of Retriever Indexing

Table 15. Ablation study on retriever index design. T and O denote type and object, respectively. All incorporates timestamp, type, object, and website.

Scheme	Key	P_{pref}	Intent
On-demand	T + O	0.677	0.542
	All	0.480	0.307
Pre-execution	T + O	0.737	0.690
	All	0.703	0.677

To validate our index design choice in retriever module, we conduct an ablation study comparing alternative indexing schemes. The original retriever extracts only *type* and *object* attributes from each history entry to form simplified keys. Following common practice in information retrieval (Wu et al., 2025; Zhong et al., 2024; Maharana et al., 2024), an alternative design incorporates all four attributes (*timestamp*, *type*, *object*, and *website*). Results on Qwen3-80B-Instruct show that the condensed indexing scheme achieves higher preference and intent scores across both access schemes. Adding timestamp and website attributes introduces noise rather than useful signal. Concatenating all attributes obscures the core semantic content in the embedding, and semantic similarity-based retrievers cannot meaningfully leverage temporal or locational information.

G. Prompt for Data Construction

Tables 16, 17, 18, and 19 detail the prompts used in our data construction pipeline. These prompts guide the generation of domain preferences, daily life event seeds, action decompositions, and personalized user history entries. Each prompt is carefully designed to enforce realism, internal consistency, and alignment with user preferences and behavioral patterns, forming the foundation of our *clarify-to-personalize* query design.

H. Examples for History Access Schemes

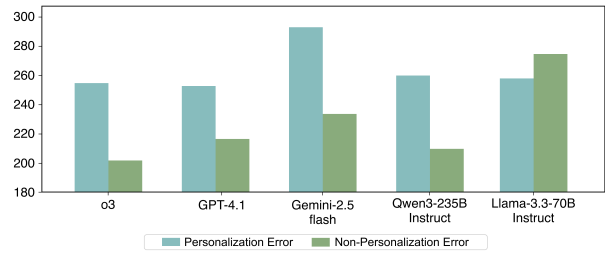
Tables 20 and 21 present execution logs illustrating the two history access schemes considered in our benchmark. In the pre-execution scheme (Table 20), the agent retrieves and reasons over multiple aspects of user history (e.g., preferred website, typical category, and format) before initiating any navigation actions. This enables the agent to construct a fully specified personalized objective prior to execution, after which navigation proceeds without further history access.

In contrast, the on-demand scheme (Table 21) interleaves history access with navigation. The agent initially retrieves only coarse-grained preferences (e.g., preferred platform) to start execution, and later queries user history again when

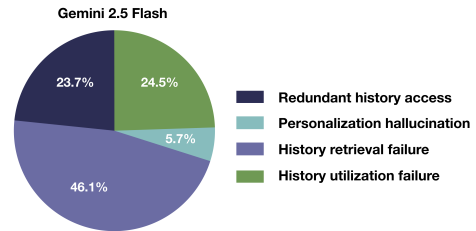
additional constraints (e.g., budget limits or course duration) become necessary. This reactive access pattern allows the agent to refine its plan during execution but requires correctly identifying when and which personalization signals should be retrieved.

Together, these examples highlight that pre-execution emphasizes upfront personalization and planning, while on-demand prioritizes adaptive personalization during execution. The differing strengths and requirements of these schemes help explain why optimal history access strategies vary across agent architectures.

I. Error Statistics



(a) All error types



(b) Personalization-related error types

Figure 5. Error statistics for AgentOccam across backbone models (top) and a breakdown of personalization-related errors for Gemini 2.5 Flash (bottom).

To investigate the sources of agent failures, we conduct an error analysis based on the main results. Errors are categorized into personalization-related errors, indicated by imperfect preference or website scores, and non-personalization-related errors, indicated by imperfect intent scores.

Figure 5(a) reports the distribution of personalization-related and non-personalization-related errors for AgentOccam with various backbone models. Across all backbones, personalization-related errors constitute a substantial portion of total failures, indicating that incorrect or incomplete use of user history remains a dominant challenge even when agents successfully navigate the web. While the overall error counts vary by model, this pattern suggests that improvements in general web navigation alone are insufficient to address personalization failures (Kim et al., 2026).

Figure 5(b) further breaks down personalization-related errors for Gemini 2.5 Flash. Among these, history retrieval failure is the most prevalent error type, accounting for nearly half of personalization errors, followed by history utilization failure and redundant history access. In contrast, personalization hallucination occurs less frequently. This breakdown highlights that personalization failures more often arise from difficulties in identifying and applying relevant historical information, such as opinions embedded in user reviews (Heo et al., 2025), rather than fabricating nonexistent preferences. Together, these findings reinforce the need for improved mechanisms that support effective history reasoning and utilization in personalized web agents.

J. Case Studies of Error Types

We also present representative execution logs for each of the four personalization error types identified in our analysis: Redundant History Access (Table 22), Personalization Hallucination (Table 24), History Retrieval Failure (Table 23), and History Utilization Failure (Table 25). These examples complement the main analysis by illustrating how each error type manifests during agent execution.

Table 16. Prompt for Domain Preference Generation

Prompt for Domain Preference Generation
Travel / Accommodation / Transportation & Restaurant Reservations
What the fields must capture
Preference: include 4-5 concise preferences regarding restaurant reservations - MUST provide user context in a concise manner. - Favorite cuisine types (e.g., Japanese, Mexican, vegetarian-friendly) - Price tiers in USD or symbols (e.g., "average meal $\leq$ \$40 per person")
Patterns: include 2-3 realistic, repeating habits such as: - "Uses OpenTable when booking business dinners during weekdays" - "Browses ExploreTock before special occasions"

Table 17. Prompt for Event Seed Generation

Prompt for Event Seed Generation - High Frequency
Generate realistic daily life events for month month of year based on the following user profile and requirements.
CRITICAL INSTRUCTIONS - All events MUST align with the provided user profile - The 'frequency related to' field in the user profile contains frequency information for each event category - strictly follow these frequencies - The 'selected subdomains' field in the user profile includes preference information about the user's tastes and lifestyle - faithfully incorporate these preferences - All events must be realistic and plausible with natural flow and timing - Keep descriptions general and avoid overly specific details - Avoid specific details that cannot be inferred from the profile (website names, brand names, etc.)
CANCELLATION RULES - Approximately 10% of cancellable events should be marked as cancelled - Add "(cancelled)" to events that are cancelled - Only cancel events where cancellation makes logical sense
User Profile {user profile}
Event Generation Guidelines
Core Categories:
Food & Dining - Create variety: restaurant visits, cafe visits, brunch meetings, dinner with friends/colleagues
Shopping - Online shopping ONLY - no offline shopping events - Include various product categories based on user preferences - Keep descriptions at appropriate level: NO more specific details
Work & Education - First identify user's occupation from profile: student, office worker, or freelancer - This determines the base routine: - Students: school attendance, study sessions, exams - Office workers: commute - Freelancers: client work, flexible schedule, networking - Include relevant special events: - Work: client meetings, conferences, presentations - Education: exams, competitions, school events, study groups - Development: online courses, workshops, certifications, skill training

Table 18. Prompt for Action Decomposition

Prompt for Action Decomposition
Decompose the following event into a sequence of plausible actions with timestamps. Event Date: {event_date} Event Description: event_description CRITICAL CONSTRAINTS: Only these 5 action types are allowed: 1. web search - searching for information online 2. web visit - visiting specific websites 3. purchase - buying products or services 4. booking - making reservations or appointments 5. review & rating - writing reviews DO NOT use any other action types. DECOMPOSITION GUIDELINES: The number of actions must reflect realistic human behavior. Don't force simple tasks to have many steps, and don't oversimplify naturally complex processes. Simple Events (1-2 actions): - Quick, single-purpose tasks - No research or comparison needed Complex Events (3-8 actions): - Multi-stage processes requiring planning - Involve research, comparison, decision-making - planning trips, organizing events, major purchases CORE REQUIREMENTS: 1. Temporal Logic Actions must occur in before/during/after timeframes: - Before: preparation actions (search, review reading, booking) - During: execution actions (purchase, web visit for real-time info) - After: follow-up actions (writing reviews, related purchases) TIMESTAMP GENERATION RULES: - Generate timestamps around the event date - Use realistic times (avoid 2-6 AM unless necessary) - Format: YYYY-MM-DD HH:MM:SS - Actions should be in chronological order 2. Action Diversity Guidelines (DO NOT copy exactly) - Shopping events: Not just 'purchase' - Include: product search → review reading → price comparison → purchase → review writing - Travel events: Full journey - Include: destination search → booking → travel info lookup → review writing - Dining events: Complete experience - Include: 'web search' (restaurant) → 'web visit' (read reviews) → 'booking' (reservation) → 'web visit' (location map) → 'review & rating' (write reviews)

Table 19. Prompt for User History Generation

Prompt for User History Generation

Personalize the following action into a detailed user history entry based on the user’s domain profile.

Action Information:

- Timestamp: {timestamp}
- Action: {action}
- Domain: {domain}
- Subdomain: {subdomain}

User’s Domain Profile:

{domain_profile}

CRITICAL REQUIREMENTS:

1. MUST reference the "preferences" field in the profile for specific brands, sizes, websites, artists, and other preferences
2. MUST consider the "patterns" field to understand user’s routine behaviors and habits
3. Generate highly specific and concrete details that align with these preferences and patterns
4. The personalization should authentically reflect this specific user’s tastes and behaviors

PERSONALIZATION GUIDELINES:

OUTPUT FORMAT:

```
{
  "timestamp": "YYYY-MM-DD HH:MM:SS",
  "type": "web search, web visit, purchase, booking, review & rating",
  "object": "detailed description with specifics from preferences",
  "website": "bare domain of the platform"
}
```

The object field MUST be extremely specific, incorporating details from the profile’s preferences and patterns fields.

Table 20. Execution log of pre-execution scheme

Query: Find books in my usual work category in my preferred format.

Step 1 (History Access):

Personalization Query: What is the user’s preferred website for finding and purchasing books?

Retrieved History Index: 1749, 850, 854, ...

Personalization Reasoning: Based on the histories provided, the user’s preferred website for purchasing books is **Amazon**. You have made five separate book purchases, all of which were on Amazon. On November 17, 2024, you purchased ‘Continuous Discovery Habits’ by Teresa Torres. On October 16, ... In all instances where details are mentioned, you leveraged Amazon Prime for quick delivery.

Step 2 (History Access):

Personalization Query: What is the user’s usual work-related book category?

Retrieved History Index: 1893, 288

Personalization Reasoning: Based on the histories provided, no explicit record of work-related book categories. However, histories show interest in **product management** through online course searches on coursera.org, not through book purchases.

Step 3 (History Access):

Personalization Query: What is the user’s preferred format for books?

Retrieved History Index: 1593, 854, 268, ...

Personalization Reasoning: Based on the user histories provided, the user’s preferred book format is **hardcover**. All book purchases recorded in the histories are for hardcover editions. ****February 18, 2024:**** Purchased ‘Inspired: How To Create Products Customers Love’ by Marty Cagan in hardcover. ****May 30, 2024:**** Purchased a hardcover edition of ‘INSPIRED: How to Create Products Customers Love’ by Marty Cagan. ****June 11, 2024:**** Purchased two hardcover business books.

Step 4 (Navigation Action):

Objective: Enhanced task: First, go to Amazon.com. Then, search for Product Management books and filter the results to show only Hardcover editions. ...

URL: <https://www.bing.com/>

Reason: The first step of the objective is to go to Amazon.com. Since the current page is a search engine, the most direct way to navigate to Amazon.com is to type “Amazon.com” into the search bar and execute the search.

Action: type [141] [Amazon.com] [1]

Step 5 (Navigation Action):

Objective: Enhanced task: First, go to Amazon.com. Then, search for Product Management books and filter the results to show only Hardcover editions. ...

URL: <https://www.amazon.com/>

Reason: The next step in the objective is to search for “Product Management” books. I will use the search bar ‘searchbox [8]’ to input the search query “Product Management” and then proceed to click the search button to execute the search.

Action: type [8] [Product Management] [1]

Step 6 (Navigation Action):

Objective: Enhanced task: First, go to Amazon.com. Then, search for Product Management books and filter the results to show only Hardcover editions. ...

URL: <https://www.amazon.com/s?k=Product+Management...>

Reason: The objective is to filter the search results to show only “Hardcover” editions. I have identified the “Hardcover” filter link (ID ‘16553’) within the “Book Format” section of the filtering menu. Clicking this link will apply the desired filter to the search results.

Action: click [16215]

...

Step 9 (Navigation Action):

Objective: Enhanced task: First, go to Amazon.com. Then, search for Product Management books and filter the results to show only Hardcover editions. ...

URL: <https://www.amazon.com/s?k=Product+Management...>

Reason: All these steps have been successfully completed. The current page shows the filtered results as requested. Therefore, the task is finished.

Action: stop [The search for “Product Management” books on Amazon.com has been performed, and the results have been successfully filtered to show only “Hardcover” editions.]

Table 21. Execution log of on-demand scheme

Query: Find sales negotiation courses that match my typical budget and course length limits.

Step 1 (History Access):

Personalization Query: user’s preferred website for finding professional development courses.

Retrieved History Index: 1113, 611, 100, ...

Personalization Reasoning: Based on the histories provided, your preferred website for finding professional development courses is **coursera.org**. You have used coursera.org on at least nine separate occasions for various activities related to professional development, specifically focusing on sales strategy and negotiation techniques.

Step 2 (Navigation Action):

Objective: Query: Find sales negotiation courses that match my typical budget and course length limits for sales/negotiation.

URL: <https://www.bing.com/>

Reason: Based on the user’s memory, the preferred website for finding professional development courses is **coursera.org**. Therefore, the first step is to navigate to this website. I will type coursera.org into the search bar and press enter to go to the website.

Action: type [144] [coursera.org] [1]

Step 3 (Navigation Action):

Objective: Query: Find sales negotiation courses that match my typical budget and course length limits for sales/negotiation.

URL: <https://www.bing.com/search?q=coursera.org...>

Reason: I have successfully searched for the user’s preferred website. The next logical step is to click on the link that leads to the Coursera homepage to begin searching for the specified courses. I will click on the link with the text “Coursera Degrees, Certificates, & Free Online Courses” to navigate to the website.

Action: click [15096]

Step 4 (Navigation Action):

Objective: Query: Find sales negotiation courses that match my typical budget and course length limits for sales/negotiation.

URL: <https://www.coursera.org/?msockid...>

Reason: The objective is to find “sales negotiation courses”. The most direct way to do this is to use the search bar provided on the homepage. I will type “sales negotiation” into the search bar and press enter to see the available courses.

Action: type [1853] [sales negotiation] [1]

Step 5 (History Access):

Personalization Query: user’s typical budget and course length limits for sales/negotiation courses.

Retrieved History Index: 1113, 904, 495, ...

Personalization Reasoning: Based on the provided histories, you have looked for courses priced **under \$30 USD**. You have a limit for total course length of **under 10 hours**. You have a preferred weekly workload of **4-6 hours per week**.

Step 6 (Navigation Action):

Objective: Query: Find sales negotiation courses that match my typical budget and course length limits for sales/negotiation.

URL: <https://www.coursera.org/search?query=sales%20negotiation...>

Reason: I have successfully searched for “sales negotiation” courses. Now I need to filter the results to match the user’s preferences. According to the user’s memory, the course length should be under 10 hours. The Durationfilter button [5907] is the most direct way to apply this constraint. I will click on it to see the available duration options and select the one that best fits the user’s needs.

Action: click [5907]

...

Step 14 (Navigation Action):

Objective: Query: Find sales negotiation courses that match my typical budget and course length limits for sales/negotiation.

URL: <https://www.coursera.org/search?query=sales%20negotiation&productDuration=Less...>

Reason: I have successfully identified multiple courses that match the user’s specified criteria for website, duration, and budget. The course titled “Copilot for Beginners: Create a Sales Pitch Presentation” is particularly relevant as it explicitly mentions “Price Negotiation”. Since I have found suitable options, the task is complete. I will now stop and provide the answer.

Action: stop [The following sales negotiation courses on Coursera match your preferences (under 10 hours, under \$30): “Copilot for Beginners: Create a Sales Pitch Presentation”, “Design a pitch deck with Canva”, and “Create a Promotional Video using Canva”.]

Table 22. Redundant History Access Error

Query: Find and compare shore excursions for our upcoming cruise that match my usual preferences and budget.

Step 1 (History Access):

Personalization Query: user’s preferred website for booking or browsing cruise shore excursions

Personalization Reasoning: Based on the provided histories, the evidence of your preferred website for booking or browsing cruise shore excursions is your use of the **Carnival Cruises** website. ...

Step 2 (Navigation Action):

URL: <https://www.bing.com/>

Reason: Following the user’s preferences and history, the search for shore excursions should be conducted on the Carnival Cruises website. The next step is to search for the Carnival Cruises website using Bing, so I can navigate there and begin searching for shore excursions that match the user’s preferences and budget.

Action: type [144] [Carnival Cruises shore excursions] [1]

Step 3 (History Access):

Personalization Query: user’s usual shore excursion preferences

Personalization Reasoning: Based on the provided user histories, there is no direct information available regarding the user’s usual shore excursion preferences. ...

Step 4 (History Access):

Personalization Query: user’s cruise itinerary (dates and ports of call for the upcoming cruise)

Personalization Reasoning: Based on the provided user histories, there is limited information available regarding your cruise itinerary. The only relevant detail found is from a memory dated 2023-10-21, which states: you booked a small-group, low-activity shore excursion for 2 adults on Carnival Cruises departing from Port Canaveral. ...

Step 5 (History Access):

Personalization Query: user’s cruise itinerary (upcoming cruise: dates and ports of call)

Personalization Reasoning: Based on the provided user histories, the available information about your upcoming cruise is as follows: Cruise Line: Carnival Cruises; Departure Port: Port Canaveral; Excursion Details: You have booked a small-group, low-activity panoramic coach sightseeing shore excursion for 2 adults. ...

Step 6 (Navigation Action):

URL: <https://www.bing.com/search?q=Carnival+Cruises+shore+excursions...>

Reason: To fulfill the user’s objective in line with their known preferences, I should proceed to the official Carnival Cruises shore excursions page. This will allow me to browse and compare available excursions, filter for low-activity or sightseeing options, and check prices, all without needing to log in or provide personal info. This step is required before any filtering or comparison based on preferences or budget.

Action: click [17259]

Step 7 (History Access):

Personalization Query: user’s specific shore excursion preferences (activity type, activity level, price range)

Personalization Reasoning: Based on the provided user histories, several histories indicate a preference for mid-range spending (\$30–\$70 per person) and highly rated services. ...

Step 8 (History Access):

Personalization Query: user’s specific shore excursion preferences (activity type, activity level, and price range)

Personalization Reasoning: Based on the provided user histories, the relevant detail is that you previously searched for family-friendly activities near Lockeland Table in East Nashville, focusing on activities suitable for all ages. ...

Step 9 (Navigation Action):

URL: <https://www.carnival.com/shore-excursions/...>

Reason: First, I must retrieve the user’s preferred website from their memory bank to support the execution of this query, and then perform the search. I should not attempt any actions that require login, phone number, or e-mail address. To match the user’s known preferences (low-activity, sightseeing, mid-range budget) and efficiently compare excursions, I should apply the following filters: - Activity Type: Sight-seeing and Motorcoach (as panoramic coach tours are a past preference) - Activity Level: Easy (to ensure low-activity options) - Price: \$50–\$99.99 and \$100 or more (to cover mid-range and slightly higher). Applying these filters will narrow down excursions to those most relevant to the user’s patterns, allowing for an effective comparison.

Action: click [2423]

Table 23. Personalization Hallucination Error

Query: Find monitors that match my usual monitor specs, and apply my preferred condition and review threshold; sort by price low to high.

Step 1 (History Access):

Personalization Query: user’s preferred website for monitor shopping

Personalization Reasoning: Based on the provided histories, the user has visited and used both eBay and Best Buy when shopping for monitors. eBay was used for price comparison, and Best Buy was checked for open-box availability. No explicit preference for either site is stated, but Best Buy appears frequently in relation to open-box deals.

Step 2 (Navigation Action):

URL: <https://www.bing.com/>

Reason: Therefore, Best Buy is a reasonable starting point for this task. Search for “Best Buy monitors” to reach the monitor category page.

Action: type [144] [Best Buy monitors] [1]

Step 3 (History Access):

Personalization Query: user’s usual monitor specs (size, resolution, panel type, refresh rate etc.)

Personalization Reasoning: The user typically searches for 27–32” 4K monitors prioritizing high color accuracy, high refresh rate, and USB-C connectivity. No brand or panel-type information is available.

Step 4 (Navigation Action):

URL: <https://www.bing.com/search?q=Best+Buy+monitors...>

Reason: Select the “Best Buy” link ([16375]) to navigate to Best Buy’s monitor section for applying the user’s specifications and filters.

Action: click [16375]

Step 5 (History Access):

Personalization Query: user’s preferred condition and minimum review threshold for monitors

Personalization Reasoning: Based on the provided histories, records indicate interests in 4K monitor and purchasing monitors under open-box condition. However, there is no explicit mention of any minimum review threshold for monitor selection.

Step 6–13 (Navigation Actions):

The agent sequentially selected “United States”, navigated to “4K Monitors”, applied the “26.5”–31.4” size filter”, and then the “Open-Box” condition filter. Finally, the results were sorted by “Price Low to High”. These steps remain consistent with the retrieved specifications and History-grounded reasoning.

Step 14 (Navigation Action):

URL: <https://www.bestbuy.com/site/searchpage...>

Reason: “The objective is to find monitors that match the user’s usual specs (4K, 27–32”, open-box, high refresh rate, USB-C, high color accuracy) and apply the preferred condition and review threshold. Since the user likely wants well-reviewed products, **I should apply the “4 Stars & Up” filter.**”

Action: click [4499]

Step 15 (Navigation Action):

URL: <https://www.bestbuy.com/site/searchpage...>

Reason: The results now display open-box 4K monitors (27–32”, USB-C, well-reviewed, sorted by price). Representative examples include LG 27” IPS UHD 4K (\$216.99, 4.8★) and Dell 27” 4K (\$239.99, 4.7★).

Action: stop

Table 24. History Retrieval Failure Error

Query: Check the current standings and this week’s schedule for my NFL team and my MLB team.
Step 1 (History Access): <i>Personalization Query:</i> What is the user’s favorite or preferred NFL team? <i>Personalization Reasoning:</i> Based on the provided user histories, the user’s favorite or preferred NFL team is the Baltimore Ravens.
Step 2 (History Access): <i>Personalization Query:</i> What is the user’s favorite or preferred MLB team? <i>Personalization Reasoning:</i> Based on the provided user histories, there is no explicit mention of the user’s favorite or preferred MLB team.
Step 3 (History Access): <i>Personalization Query:</i> What is the user’s preferred website for checking sports standings and schedules? <i>Personalization Reasoning:</i> Based on the provided user histories, the user’s preferred website for checking sports standings and schedules is Fox Sports (foxsports). If the user uses any other sports websites, this is not reflected in the current histories.
Step 4 (Navigation Action): <i>URL:</i> https://www.bing.com/ <i>Reason:</i> To proceed according to the user’s preferences, I need to go to the Fox Sports website. Since there is no direct link, the most efficient step is to use the search box to search for “Fox Sports Baltimore Ravens standings and schedule” to directly reach the relevant section on Fox Sports. <i>Action:</i> type [141] [Fox Sports Baltimore Ravens standings and schedule] [1]
Step 5 (Navigation Action): <i>URL:</i> https://www.bing.com/search?q=Fox+Sports+Baltimore+Ravens+standings+and+schedule... <i>Reason:</i> The user’s preferred source is Fox Sports, and the top links ([17517]/[17528] and [17794]) are directly relevant to the Baltimore Ravens’ standings and schedule. To fulfill the objective, I should proceed to the Fox Sports page for the Baltimore Ravens’ schedule and standings. Link [17794] appears to be the most comprehensive for both the schedule and scores, which likely includes this week’s schedule and potentially a link to standings or the standings themselves. <i>Action:</i> click [17794]
Step 6 (Navigation Action): <i>URL:</i> https://www.bing.com/search?q=Fox+Sports+Baltimore+Ravens+standings+and+schedule&form... <i>Reason:</i> The objective is to retrieve both the current standings and this week’s schedule for the Baltimore Ravens from Fox Sports. The most direct links to Fox Sports are [17517], [17528] (standings) and [17777] (schedule). Since the previous click on [17794] did not navigate away from Bing, the next best action is to click [17777], which is another direct Fox Sports link for the Baltimore Ravens’ schedule and may provide the required information. <i>Action:</i> click [17777]
Step 7 (Navigation Action): <i>URL:</i> https://www.foxsports.com/nfl/baltimore-ravens-team-schedule <i>Reason:</i> The current page provides both the Ravens’ current standings and their schedule, including this week’s game. The record is explicitly stated at the top, and the upcoming games table includes this week’s schedule. No information about an MLB team can be provided due to lack of user preference. All required information is now gathered and ready to be summarized for the user. <i>Action:</i> stop [Baltimore Ravens (NFL) on Fox Sports: - Current standing: 1-3, 3rd in AFC North. - This week’s game: vs. Houston Texans, Monday 10/5, 2:00AM at M&T Bank Stadium, Baltimore, MD (CBS). - No MLB team schedule or standings are provided, as no preferred MLB team is identified in your histories.]

Table 25. History Utilization Failure Error

Query: Check seat availability for upcoming concerts in my preferred genre with seats in my preferred location.

Step 1 (History Access):

Personalization Query: What is the user’s preferred website for buying concert tickets?

Personalization Reasoning: Based on the user histories provided, SeatGeek is the only website explicitly mentioned in the context of checking ticket availability.

Step 2 (History Access):

Personalization Query: What is the user’s preferred music genre?

Personalization Reasoning: Based on the user histories, there is no explicit mention of a preferred music genre. However, History indicates an interest in specific artists such as Kendrick Lamar and SZA, for whom the user previously searched concert venues.

Step 3 (History Access):

Personalization Query: What is the user’s preferred seating location within a concert venue?

Personalization Reasoning: The user’s preferred seating location within a concert venue is in the lower-bowl sections near the stage.

Step 4 (History Access):

Personalization Query: What is the user’s home city or the city they are interested in for finding concerts?

Personalization Reasoning: Based on the user histories, the city the user is interested in for concerts is **Los Angeles**.

Step 5 (Navigation Action):

URL: <https://www.bing.com/>

Reason: The first step of the task is to navigate to SeatGeek.com. Since the current page is a search engine, the most direct way to do this is to type “SeatGeek.com” into the search bar and press Enter.

Action: type [141] [SeatGeek.com] [1]

Step 6 (Navigation Action):

URL: <https://www.bing.com/search?q=SeatGeek.com...>

Reason: I have successfully searched for SeatGeek.com. The next step is to click the first search result, which leads to the SeatGeek homepage.

Action: click [15414]

Step 7–8 (Navigation Actions):

The agent searched sequentially for “Kendrick Lamar” and “SZA” concerts on SeatGeek. However, these searches were conducted at the global level **without applying the retrieved city constraint (Los Angeles)**, thereby overlooking the user’s explicit location preference during reasoning and action selection.

Step 9 (Navigation Action):

URL: <https://seatgeek.com/sza-tickets>

Reason: According to SeatGeek, there are no upcoming concerts in Los Angeles for either Kendrick Lamar or SZA. While the website mentions that SZA will be an opening act for Kendrick Lamar’s 2025 Grand National Tour, no specific dates or ticket listings are currently available. The agent stops here without verifying lower-bowl seat availability or applying the city constraint.

Action: stop [No upcoming concerts found for Kendrick Lamar or SZA in Los Angeles. Seat availability could not be checked.]
